

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         double[] data = {
8             115, 182, 191, 31, 196, 1099, 5, 172, 10, 179,
9             83, 21, 20, 21, 186, 177, 195, 193, 188, 199,
10            62, 109, 105, 183, 110
11        };
12
13        Array.Sort(data);
14        int n = data.Length;
15
16        Console.WriteLine("Sorted Data:");
17        for (int i = 0; i < n; i++)
18            Console.Write(data[i] + " ");
19
20        Console.WriteLine("\n");
21
22        //MEAN
23        double sum = 0;
24        for (int i = 0; i < n; i++)
25            sum += data[i];
26
27        double mean = sum / n;
28        Console.WriteLine("Mean = " + mean);
29
30        //MEDIAN
31        double median;
32        if (n % 2 == 0)
33            median = (data[n / 2] + data[n / 2 - 1]) / 2;
34        else
35            median = data[n / 2];
36
37        Console.WriteLine("Median = " + median);
38
39        //MODE
40        int maxCount = 0;
41        double mode = data[0];
42
43        for (int i = 0; i < n; i++)
44        {
45            int count = 0;
46            for (int j = 0; j < n; j++)
47            {
48                if (data[i] == data[j])
49                    count++;
```

```
50     }
51
52     if (count > maxCount)
53     {
54         maxCount = count;
55         mode = data[i];
56     }
57 }
58
59 Console.WriteLine("Mode = " + mode);
60
61 //VARIANCE
62 double varSum = 0;
63 for (int i = 0; i < n; i++)
64     varSum += Math.Pow(data[i] - mean, 2);
65
66 double variance = varSum / n;
67 Console.WriteLine("Variance = " + variance);
68
69 //STANDARD DEVIATION
70 double std = Math.Sqrt(variance);
71 Console.WriteLine("Standard Deviation = " + std);
72
73 //QUARTILES
74 double q1 = data[n / 4];
75 double q2 = median;
76 double q3 = data[3 * n / 4];
77
78 Console.WriteLine("Q1 = " + q1);
79 Console.WriteLine("Q2 = " + q2);
80 Console.WriteLine("Q3 = " + q3);
81
82 //PERCENTILES
83 Console.WriteLine("P20 = " + data[(int)(0.2 * n)]);
84 Console.WriteLine("P50 = " + median);
85
86 // RANGE
87 double range = data[n - 1] - data[0];
88 Console.WriteLine("Range = " + range);
89
90 //IQR
91 double iqr = q3 - q1;
92 Console.WriteLine("IQR = " + iqr);
93
94 //SUM OF DEVIATIONS
95 double sumDev = 0;
96 for (int i = 0; i < n; i++)
97     sumDev += (data[i] - mean);
98
```

```
99         Console.WriteLine("Summation of Deviations = " + sumDev);
100
101         //OUTLIERS
102         double lower = q1 - 1.5 * iqr;
103         double upper = q3 + 1.5 * iqr;
104
105         Console.WriteLine("\nOutliers:");
106         for (int i = 0; i < n; i++)
107         {
108             if (data[i] < lower || data[i] > upper)
109                 Console.WriteLine(data[i]);
110         }
111
112         //CHECK EACH VALUE
113         Console.WriteLine("\nCheck each value:");
114         for (int i = 0; i < n; i++)
115         {
116             if (data[i] < lower || data[i] > upper)
117                 Console.WriteLine(data[i] + " -> Outlier");
118             else
119                 Console.WriteLine(data[i] + " -> Normal");
120         }
121     }
122 }
```